W.H. Croxford High School · Mathematics Department

Mathematics 15 Course Outline

2026–27
5 credits

Teacher	Email	Google Classroom	Class website
Mr. Ryan Cathcart Room 2997	**rcathcart@rvschools.ab.ca** Students and parents welcome	**Code given in class** Website link, files, announcements	**rmcathcart-lab .github.io/math15** Lessons, practice, calendar, tests

About the course

Math 15 (*Competencies in Math 15*) is a locally developed course that gets you ready for Math 10C. It rebuilds number sense and then works through the skills Math 10C is made of: factors and roots, exponents, polynomials and factoring, relations and functions, linear relations and systems of equations, at a pace that leaves room to practise.

Math 15 → **Math 10C** → **Math 20-1** or 20-2
Math 15 → **Math 20-3** → **Math 30-3**

Math 15 comes after Math 9 and is worth 5 credits. Students who complete it go on to Math 10C or Math 20-3. A high school diploma needs 100 credits, including a 20-level math.

Success in math is mostly persistence. Everyone in this room can do well in this course by doing the practice, asking questions early and learning from mistakes. Wrong answers are where the learning happens.

A typical class

1. **Arrive ready**: laptop charged, calculator, pencil and paper on your desk.
2. **Warm-up** on the board while attendance is taken.
3. **Lesson**: I teach from the class website. Take notes on paper, or print the notes booklet for the unit.
4. **Practice** on the website: new numbers every time, hints when you're stuck, and your level on each outcome.
5. **Exit slip** almost every day: a short check on today's outcome, written on the website.

Classes run about 88 minutes, bell to bell. The class calendar on the website shows each day's lesson and every test date, and it is always up to date.

The course at a glance

Unit	Topic	Lessons	Outcomes
1	Number	8	AN1 · AN2
2	Exponents	10	AN3
3	Polynomial Operations	7	AN4
4	Factoring Polynomial Expressions	10	AN5
5	Relations and Functions	12	RF1 · RF2 · RF4 · RF5 · RF8
6	Characteristics of Linear Relations	10	RF3
7	Equations of Linear Relations	10	RF3 · RF4 · RF5 · RF6 · RF7
8	Systems of Linear Equations	8	RF9

75 lessons and a unit test at the end of every unit. Dates for your class are on the class calendar.

What you will learn: the 14 outcomes

Math 15 uses these outcomes from Alberta Education's Mathematics 10C Program of Studies (2008), with expectations set for an introductory course. These outcomes are what you are assessed on.

Algebra and Number

AN1	Demonstrate an understanding of factors of whole numbers by determining the: prime factors; greatest common factor; least common multiple; square root; cube root.
AN2	Demonstrate an understanding of irrational numbers by: representing, identifying and simplifying irrational numbers; ordering irrational numbers.
AN3	Demonstrate an understanding of powers with integral and rational exponents.
AN4	Demonstrate an understanding of the multiplication of polynomial expressions (limited to monomials, binomials and trinomials), concretely, pictorially and symbolically.
AN5	Demonstrate an understanding of common factors and trinomial factoring, concretely, pictorially and symbolically.

Relations and Functions

RF1	Interpret and explain the relationships among data, graphs and situations.
RF2	Demonstrate an understanding of relations and functions.
RF3	Demonstrate an understanding of slope with respect to: rise and run; line segments and lines; rate of change; parallel lines; perpendicular lines.
RF4	Describe and represent linear relations, using: words; ordered pairs; tables of values; graphs; equations.
RF5	Determine the characteristics of the graphs of linear relations, including the: intercepts; slope; domain; range.
RF6	Relate linear relations expressed in: slope–intercept form ($y = mx + b$); general form ($Ax + By + C = 0$); slope–point form ($y - y_1 = m(x - x_1)$) to their graphs.
RF7	Determine the equation of a linear relation, given: a graph; a point and the slope; two points; a point and the equation of a parallel or perpendicular line to solve problems.
RF8	Represent a linear function, using function notation.
RF9	Solve problems that involve systems of linear equations in two variables, graphically and algebraically.

How you are assessed

Math 15 uses **outcomes-based assessment**. Your mark shows which outcomes you can do and how well, not how many points you collected. For each outcome you earn an **indicator**. On a unit test, it reads straight from three boxes of questions:

Beginning box	Progressing box	Mastery box
the core skill	the skill in typical questions	harder, multi-step and new situations

A box is **secure** when you earn ¾ of its points and **half done** at ½:

MAS	**Mastery**	Beginning, Progressing and Mastery boxes all secure.	**85–100%**
ADV	**Advancing**	Beginning and Progressing secure; at least half of Mastery.	**75–85%**
PRG	**Progressing**	Beginning and Progressing secure.	**65–75%**
EMG	**Emerging**	Beginning secure; at least half of Progressing.	**55–65%**
BEG	**Beginning**	Beginning box secure.	**45–55%**
LIM	**Limited**	At least half of the Beginning box.	**25–45%**
NM	**Not Meeting**	Less than half of the Beginning box.	**0–25%**
IE	**Insufficient Evidence**	Not enough work yet to judge, such as a missed test not yet made up.	**—**

From outcomes to your report card

Part marks and unit grades

Questions can earn part marks: a small slip keeps ⅔ of the point, and a bigger but reasonable mistake keeps ⅓. Your **unit grade** is the average of the unit's outcomes. Exit slips check one or two outcomes from the day's lesson, and they are graded the same way as a unit test: by their Beginning, Progressing and Mastery boxes.

Your final mark

Your report card mark comes from your grades on the 14 outcomes. Everything you do **in class** adds evidence for the outcomes it covers: unit tests, exit slips and practice in class. When I make a judgment call on an outcome, I use what I've seen you do in class.

Tests and exit slips

Unit tests and exit slips are written **on the class website**, in class, on your laptop. They pop up full screen when I launch them, and every student gets their own numbers. Tests end at the end of class. **Leaving the test window, switching tabs or leaving full screen submits your work and locks the test**; only I can unlock it.

Once I release the results, you see your answers, the correct answers and a worked solution for each question.

Missed a test?

Make it up at the **Test Centre** in the Learning Commons, Fridays 2:15–3:45. Fill out the missed-assessment form first. If you miss your Test Centre time, you lose that chance.

Practice and due dates

Each lesson's practice is on the website, with a due date on the class calendar. Due dates set the pace, so you're ready for the exit slip and the unit test. Practice isn't marked down for being late, since marks come from work in class, but unfinished practice matters for retests.

Retests: earn another try

A retest is earned with practice, and your **best result counts**, so a retest can only help.

1. **Do the practice.** Finish every required question in the unit without seeing the solution. (Saw one? Redo a new version without it.)
2. **Request a retest.** The button appears on that unit on your home page.
3. **Write it in class.** I open a retest just for you, with new numbers, while the class works.

Want another retest after that? Finish **all the extra practice** in the unit first.

> **Badges.** Every outcome has a badge beside your name on the website, from Pattern (Not Meeting) up to Master (Mastery). It shows the best level you have reached from practice in class or a unit test.

What to bring every day

- **Laptop or Chromebook**, charged. Croxford is a bring-your-own-device school; ask the office about borrowing a Chromebook for the year.
- **Scientific calculator** like the TI-30XIIS (about $15–20). Not your phone. A few loaners are available in class, and a TI-30XIIS is built into the class website.
- **Pencil, eraser** and **paper or a notebook** for notes (or your printed notes booklet).

Calculators

Many Math 15 units are **no-calculator** units: number sense, exponents, polynomials and factoring are skills to have in your head. A test or exit slip always says whether a calculator is allowed. When one is, it follows Alberta Education's rules: a calculator is **not allowed** if it gives exact answers such as 3√2 instead of decimals, stores programs or notes, or has anything wireless (for example the Casio fx-991 or TI-30XS MultiView). Not sure about yours? Bring it in and we'll check it.

Expectations

- **Be on time and ready** to start when the bell goes.
- **Phones off and away** during class (the provincial rule). If you leave the room, your phone stays with me; take the hall pass, one person at a time.
- **Compassionate Cavs:** be kind, encourage each other, and listen when someone is teaching. This is a room where you can be wrong out loud.
- **Own your learning.** Missed a class? The lesson, notes and practice are on the website: catch up before the next class and ask me about anything unclear.

Getting help

Asking early is the biggest predictor of doing well. Ask during work time in class (that's what it's for), use the hints in the practice, or come see me at lunch or before or after school. The only time I won't help is during a test.

Academic integrity

Every assessment must show **your own** thinking. On tests and exit slips, any of these is cheating:

- talking to or communicating with anyone, by any means;
- using notes, devices, websites or apps that weren't allowed;
- using **AI** or photo-solver apps for any part of the work;
- sharing questions or answers with students who haven't written yet.

On practice, copying someone's work costs you the practice. AI and solver apps are fine for *checking* your work, not for producing it.

When academic dishonesty happens, I will talk with you and contact home. It can mean a 0 on that assessment, or an alternate assessment of the outcome at my discretion. Serious or repeated cases are referred to administration.

For parents and guardians

Marks are posted in **PowerSchool** by outcome: an indicator for each outcome a unit test or exit slip has covered, plus each unit grade.

For the day-to-day picture, ask your student to show you their home page on the class website: their level on every outcome, their badges, what's due and the next test date.

This course follows Rocky View Schools **AP 360**, Assessment and Communication of Student Learning, and **AP 223**, Student Reassessment (rockyview.ab.ca), and the Croxford student handbook. Questions about this outline? Email rcathcart@rvschools.ab.ca.